



AI Technicians Research Project

Enhancing AI Workforce Readiness with Stakeholder Focused Capstone Projects



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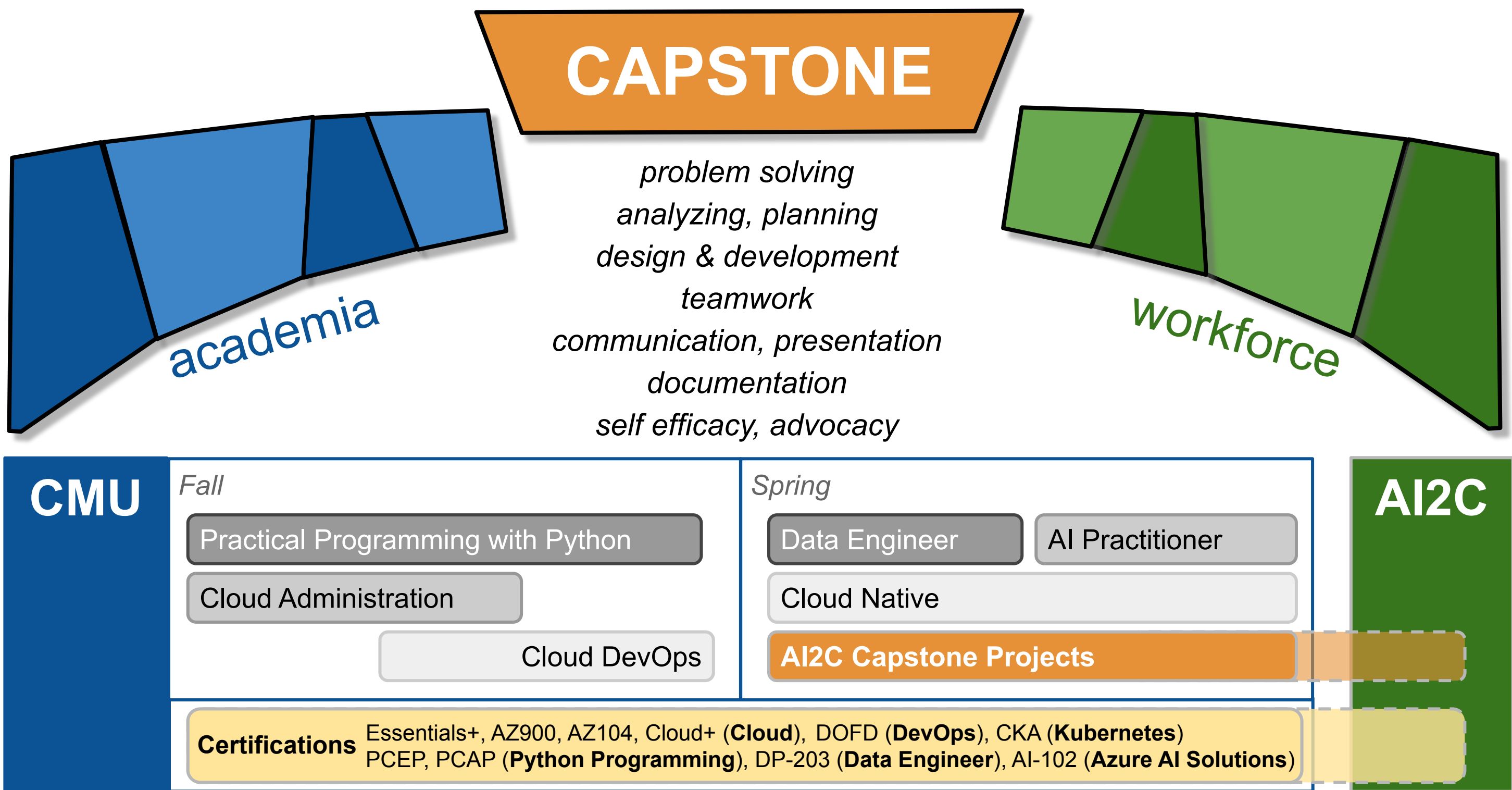
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Research Problem

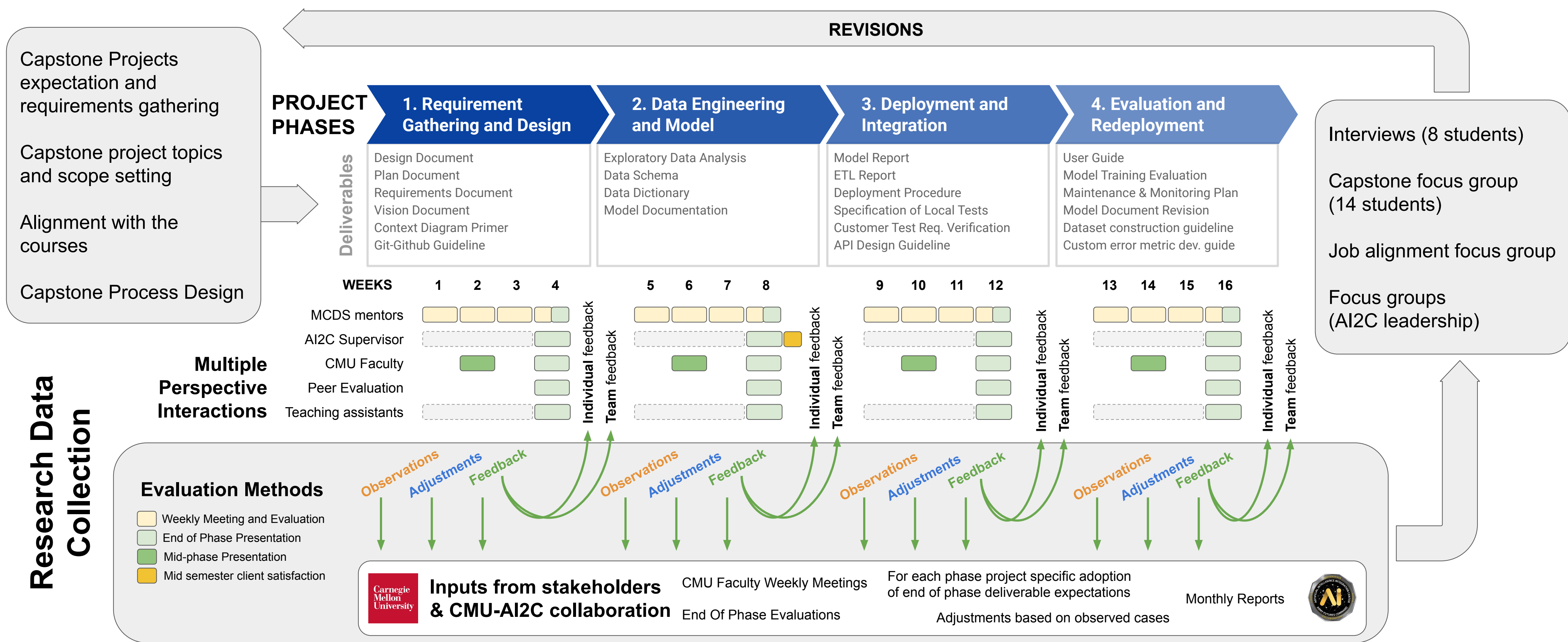
How can we ensure that capstone projects bridge to the workforce and enhance the necessary skills of participants?



Approaches

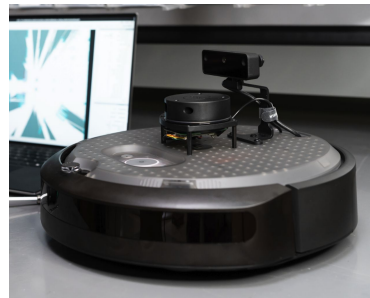
- Constant continuous measurement
- Frequent and actionable feedback
- Multiple perspective
 - CMU Faculty
 - AI2C Supervisors
 - MSDC Mentors
 - Teaching Assistants
 - Peers

AI Technicians Program



AI Technicians 2023-2024 Capstone Projects

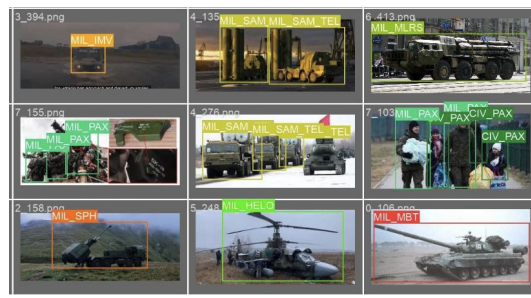
Spy Turtle
Robotics and AI in surveillance



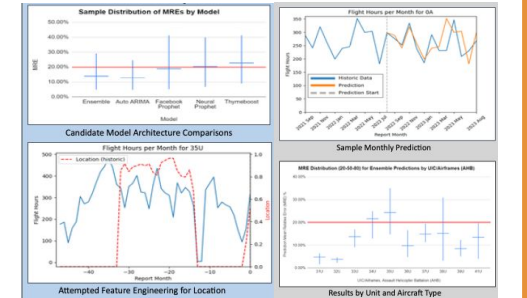
MIMIR-V
Scaling LLM Services for the U.S. Army



Elucidate
Object detection as a service

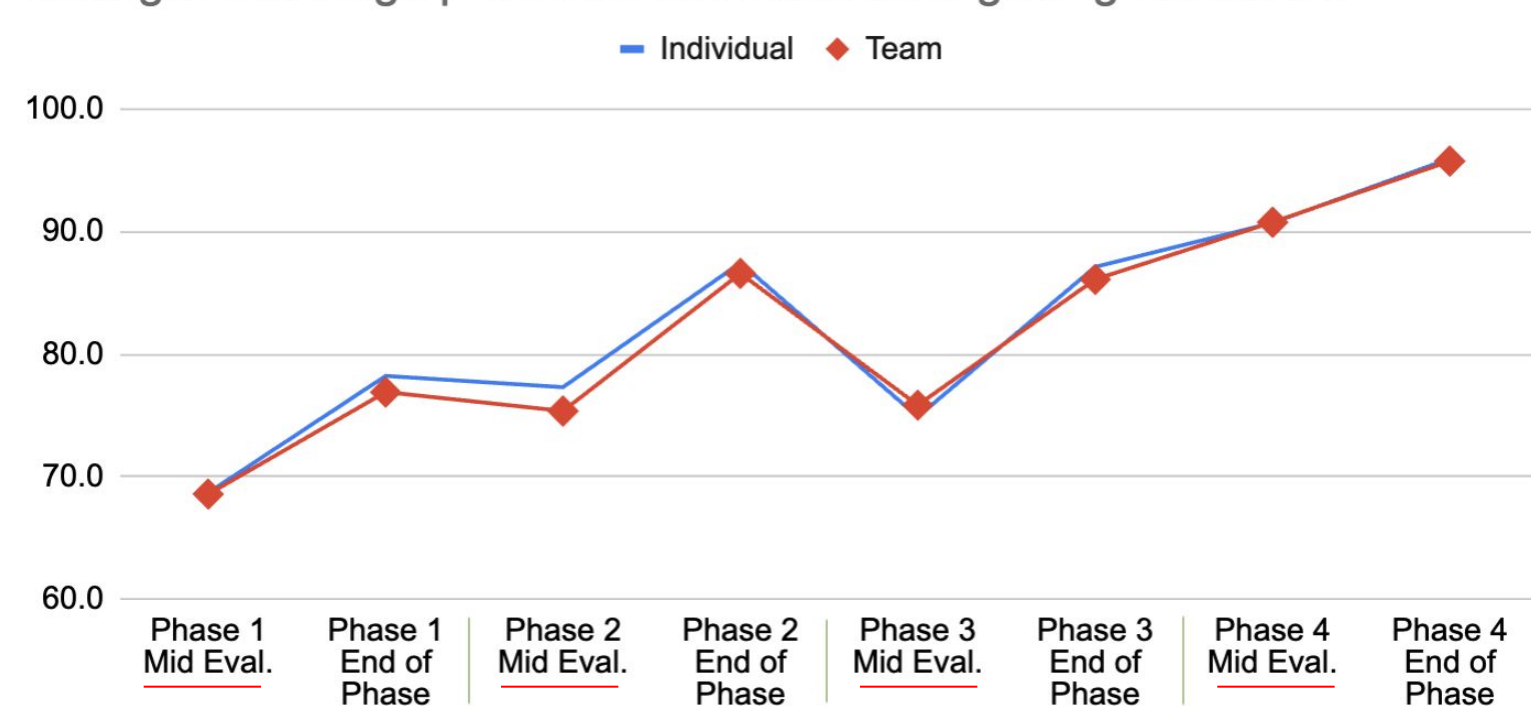


Griffin
Flight hour predictor

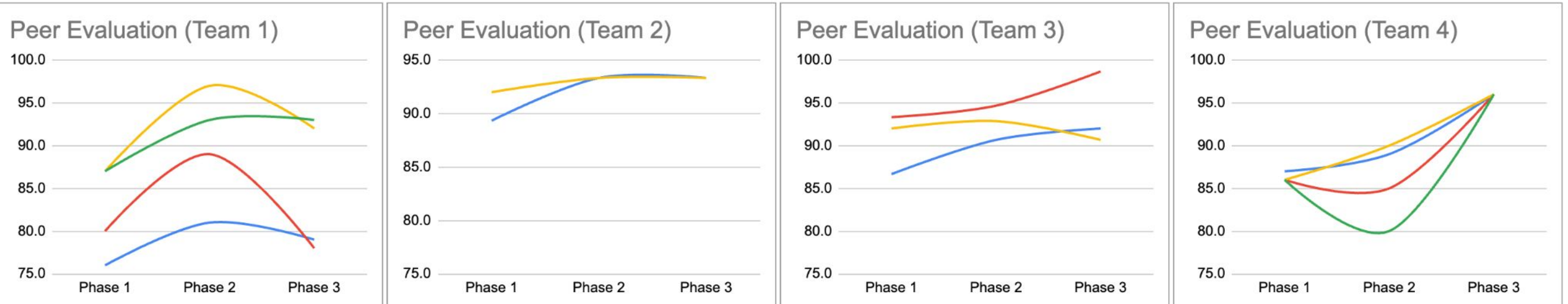


Results and Observations

Change in average presentation scores after getting feedbacks

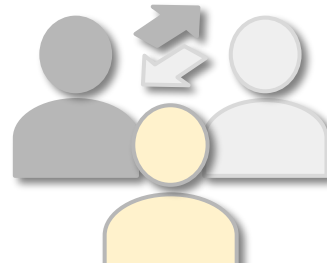


- Frequent actionable feedback** help students to improve skills in multiple aspects
- Well established data infrastructure** helps efficient
 - monitoring which is important to provide timely feedback.
 - communication between multiple stakeholders.
 - data collection and analysis from multiple sources.



Repeated **peer evaluations** help understanding team dynamics and supports providing individual feedback.

- "There is definitely a language differential, between how the military communicates with each other, and how academics communicates with each other."
- "Mentor's priority is production ASAP. CMU wants experimentation."
- "CMU wants more than what AI2C wants"



Multiple perspective exposure helps balancing skills and provides transition of various expectations to the outcome.

- "No clue how to do certain things." ⇒ "I'm fully confident in ability to do devops as career"
- "... where we're at now, like I guess I'm as confident as I can be after 6 months."
- "I have been very impressed with the program, and just see, like looking back on August before I started to. ... if I sit back and think about it all, it's kind of mind blowing. ... it's pretty impressive."

Capstone as a practical part of program seems to support **self efficacy**.

Gaps and issues to consider (both academy & workforce)

Clarified shared expectations.

Adding lecture or activities to improve soft skills.

Course workload and capstone balance.

Course topic and capstone project alignments.